CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 1: Real Numbers

Previous year board questions, topic-wise: 20 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 HCF and LCM by prime factorisation

1. Find the HCF of 960 and 432. (a) 48 (b) 54 (c) 72 (d) 36 CBSE 2026, 1 mark
2. Let m be the HCF and n the LCM of 98 and 28. Find the value of $n - 7m$. CBSE 2025, 1 mark
3. Two positive integers are $p = 18a^2b^4$ and $q = 20a^3b^2$, where a and b are primes. What is the LCM of p and q? (a) $2a^2b^2$ (b) $180a^2b^2$ (c) $12a^2b^2$ (d) $180a^3b^4$ CBSE 2024, 1 mark
4. At a workshop there are 48 French teachers, 80 Hindi teachers and 144 English teachers. Every room must hold the same number of teachers, and all the teachers in a room must teach the same subject. What is the least number of rooms needed? CBSE 2024, 3 marks
5. What is the largest number that leaves remainder 1 when it divides 85 and remainder 2 when it divides 72? CBSE 2023, 2 marks
6. Find the HCF of the smallest prime number and the smallest composite number. CBSE 2018, 1 mark
7. Find the HCF and LCM of 404 and 96, and check that HCF $\times$ LCM equals the product of the two numbers. CBSE 2018, 3 marks

2 The Fundamental Theorem of Arithmetic

8. How is the natural number 2 classified? (a) prime (b) composite (c) both prime and composite (d) neither prime nor composite CBSE 2026, 1 mark
9. For a natural number n, what is the last digit of 6^n? (a) 0 (b) 6 (c) 3 (d) 2 CBSE 2026, 1 mark
10. For which natural numbers n is $(-1)^n + (-1)^8 = 0$? CBSE 2025, 1 mark
11. Show that $5 \times 11 \times 17 + 3 \times 11$ is a composite number. CBSE 2024, 2 marks

3 Irrational numbers and their proofs

12. Show that $\sqrt{3}$ is irrational. CBSE 2026, 3 marks
13. Which of these is a rational number lying between $\sqrt{3}$ and $\sqrt{5}$? (a) $1.4142387954012\ldots$ (b) $2.32\overline{6}$ (c) π (d) 1.857142 CBSE 2025, 1 mark
14. Show that $\dfrac{1}{\sqrt{5}}$ is an irrational number. CBSE 2025, 3 marks
15. Given that $\sqrt{3}$ is irrational, prove that $5 - 2\sqrt{3}$ is irrational. CBSE 2024, 2 marks

16. Show that $\sqrt{5}$ is irrational. CBSE 2023, 3 marks

17. $\triangle ABC$ is right-angled at B with $AB = 2$ cm and $BC = 3$ cm. Assertion (A): The perimeter of $\triangle ABC$ is a rational number. Reason (R): The sum of the squares of two rational numbers is always rational. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true CBSE 2023, 1 mark

18. Write a rational number that lies between $\sqrt{2}$ and $\sqrt{3}$. CBSE 2019, 1 mark

19. Show that $\sqrt{2}$ is irrational. CBSE 2019, 3 marks

20. Given that $\sqrt{2}$ is irrational, prove that $5 + 3\sqrt{2}$ is irrational. CBSE 2018, 2 marks

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